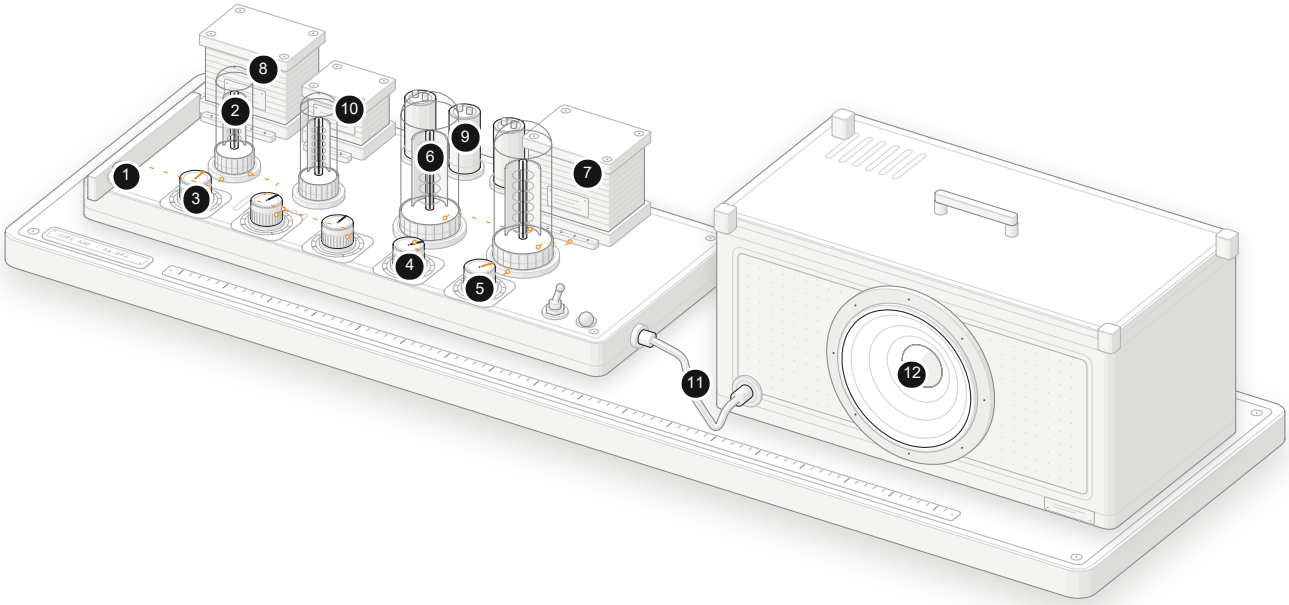
Tube guitar amp: how gain becomes crunch

AN-004



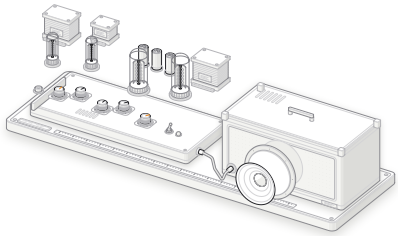
PARTS

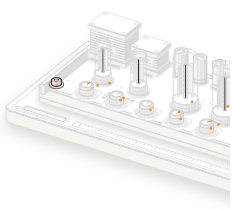
01	Input jack
02	Preamp tubes V1, V2
03	Gain knob
04	Tone knobs
05	Master knob
06	Power tubes V3, V4
07	Output transformer
08	Power transformer
09	Filter capacitors
10	Choke
11	Speaker cable
12	Speaker

SPECIFICATION

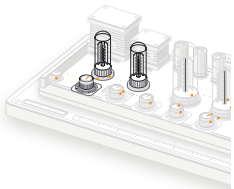
Input	0.3 V peak
Preamp V1	gain 20, clips at 40 V
Preamp V2	gain 45, clips at 80 V
Tone stack	loses 20 dB
Power tubes	gain 10, clip at 45 V
Gain pot	audio taper, (knob / 10) squared

EXPLODED

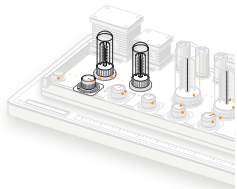




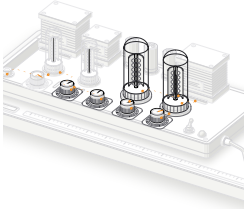
01
The pickup signal



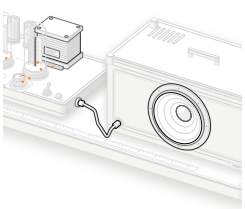
02
Preamp tubes gain it



03
V2 runs out of room



04
Power tubes push it



05
The cone follows